

Building a Quantitative Backtesting Engine from Scratch

An Independent Exploration of Trend-Following Strategies and the Limitations of Fixed Stop-Losses

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GitHub Repository: github.com/ricardo05143018-creator/quant-backtest-engine

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Abstract

This project documents my independent learning process of building a custom backtesting engine in Python. Initially, I attempted to test a basic Dual Moving Average (DMA) crossover strategy using standard vectorised pandas operations. However, after realising that this approach introduces look-ahead bias, I rebuilt the engine from scratch using a day-by-day simulation loop to ensure that each trading decision only uses information available up to the previous closing price.

I added a hard 5% fixed stop-loss and tested this framework across three markets using full-year 2025 data. The results showed a clear pattern. On AAPL (lower volatility), the strategy entered once, held for 126 days without triggering the stop-loss, and returned 31.6% (Sharpe 2.16). On TSLA (higher volatility), it entered and exited four times, averaging just 17 days per trade, and lost 5.4% while buy-and-hold gained roughly 80% over the same period. On BTC-USD, two of three positions were stopped out within weeks.

I later learned this failure mode is called the “whipsaw” effect. The central lesson is that a fixed-percentage stop-loss implicitly assumes constant volatility, which is reasonable for AAPL but fundamentally wrong for TSLA and Bitcoin. This directly establishes my next learning objective: studying time-series analysis to implement volatility-adaptive risk controls such as the Average True Range (ATR).

I acknowledge that because the MA parameters were optimised on the same data used for evaluation, the reported Sharpe Ratio for AAPL likely overstates true out-of-sample performance.

1 Introduction and Motivation

I became interested in quantitative finance after noticing that many online trading tutorials use oversimplified backtests—typically just a few lines of pandas code that calculate signals across an entire dataset at once. This felt unrealistic to me: in real life, you cannot use tomorrow’s price to make today’s decision.

The goal of this project was to build something more careful. I wanted to:

1. Construct a backtesting engine that processes data day by day, so that no future information leaks into past decisions.
2. Add a simple risk management rule (a 5% stop-loss) and observe how it behaves.
3. Test the same strategy on assets with different volatility characteristics to see whether it generalises or breaks down.

What I discovered surprised me: the stop-loss that helped on a stable stock actively destroyed performance on volatile assets. Understanding *why* this happens became the most valuable part of the project.

2 Methodology

2.1 Data Retrieval

I retrieved daily OHLCV price data for the period January–December 2025 using the `yfinance` API in Python. I used a fixed date range rather than a rolling lookback window so that all results are fully reproducible.

2.2 Signal Generation

I used `pandas`' `.rolling().mean()` function to compute a short-term and a long-term moving average of the daily closing price. A buy signal is generated when the short MA crosses above the long MA (a **Golden Cross**); a sell signal when the reverse occurs (a **Death Cross**).

This vectorised calculation is fast, but using these pre-computed signals directly to simulate trades introduces look-ahead bias: the signal value on day t already embeds that day's closing price, so a trade executed at day t 's close would implicitly use information it should not have access to.

2.3 Time-Step Simulation and Eliminating Look-Ahead Bias

To solve this, I rebuilt the core trading logic as a day-by-day `for` loop. The constraint is: **any decision on day t can only use data available at the close of day $t - 1$** . Specifically, the entry signal is read from `signals[t-1]`, the entry price is set to `closes[t-1]`, and the return captured is $(\text{closes}[t] - \text{closes}[t - 1]) / \text{closes}[t - 1]$.

This is conceptually simple, but writing it correctly required careful attention to array indexing. My first version had an off-by-one error that took me a while to find and fix.

2.4 Risk Management: The 5% Fixed Stop-Loss

Within the daily loop, if a long position has fallen 5% or more below its entry price, the engine exits immediately—regardless of what the MA signals say. I assumed execution at exactly the 5% threshold (a standing stop order). In reality, overnight price gaps can cause worse fills than this, so the assumption is slightly optimistic.

I also applied a 0.1% transaction cost on every entry and exit as a rough estimate of commissions and slippage.

2.5 Performance Metrics

I evaluated each backtest run using three levels of metrics:

- **Annualised Sharpe Ratio:** Mean daily strategy return divided by its standard deviation, scaled by $\sqrt{N}$, where $N = 252$ for equities and $N = 365$ for BTC-USD. The distinction matters: BTC trades every calendar day, so using $\sqrt{252}$ would understate annualised volatility and artificially inflate the Sharpe Ratio.
- **Maximum Drawdown (MDD):** The largest peak-to-trough decline in the cumulative strategy value.
- **Trade-level statistics:** I logged each individual trade with its entry price, exit price, holding duration in days, and exit reason (Death Cross or stop-loss). These per-trade statistics turned out to be the most informative diagnostic for understanding where the strategy failed. Note that the per-trade `return_pct` in the trade log accurately reflects stop-loss exits, but records only the exit-day commission for Death Cross exits; the daily holding-period returns are captured separately in the cumulative equity curve.

3 Results and Parameter Optimisation

3.1 Grid Search

Rather than guessing window lengths, I ran a grid search: testing all combinations of short windows $\in \{3, 5, 8, 10\}$ against long windows $\in \{20, 30, 40, 50, 60\}$ on AAPL’s 2025 data, and selecting the pair with the highest Sharpe Ratio. The best-performing combination was **MA(8, 60)**.

Important caveat: These parameters were selected and evaluated on the same dataset, so the AAPL results below are likely optimistic due to overfitting. A rigorous analysis would use a separate held-out test period. I note this as a key limitation.

3.2 Cross-Market Performance Summary

Table 1 summarises the strategy’s performance across all three markets using MA(8, 60) with a 5% stop-loss.

Table 1: Strategy performance summary (2025), MA(8, 60) with 5% stop-loss. “SL Exits” = number of stop-loss triggered exits.

Market	Return	Sharpe	MDD	Trades	Win Rate	Avg. Hold	SL Exits
AAPL	+31.6%	2.16	−5.6%	1	100%	126 days	0
TSLA	−5.4%	−0.05	−29.3%	4	0%	17 days	2
BTC-USD	+8.9%	0.58	−17.4%	3	0%	46 days	2

The pattern across the three markets directly supports the central thesis: as asset volatility increases, the fixed stop-loss is triggered more frequently, average holding periods shorten, and strategy performance deteriorates relative to buy-and-hold.

3.3 Performance on AAPL

Applying MA(8, 60) with a 5% stop-loss to AAPL produced the following results:

- **Cumulative Return:** +31.6%
- **Annualised Sharpe Ratio:** 2.16
- **Maximum Drawdown:** −5.6%
- **Trade statistics:** 1 completed trade, held for 126 days, 0 stop-loss exits.

Figure 1 shows the strategy’s equity curve against buy-and-hold. The strategy entered once on a Golden Cross, held for the majority of the year without the stop-loss being triggered, and exited at year end. Because it remained long for 126 consecutive days, its return closely tracks the underlying stock—which is the expected outcome when no premature exits occur. The position indicator in the lower panel confirms the strategy was almost entirely in a long position throughout the year.

4 Limitations and Lessons Learned: The Whipsaw Effect

While the strategy worked on AAPL, applying the exact same parameters to TSLA and Bitcoin revealed a serious structural flaw in my design.

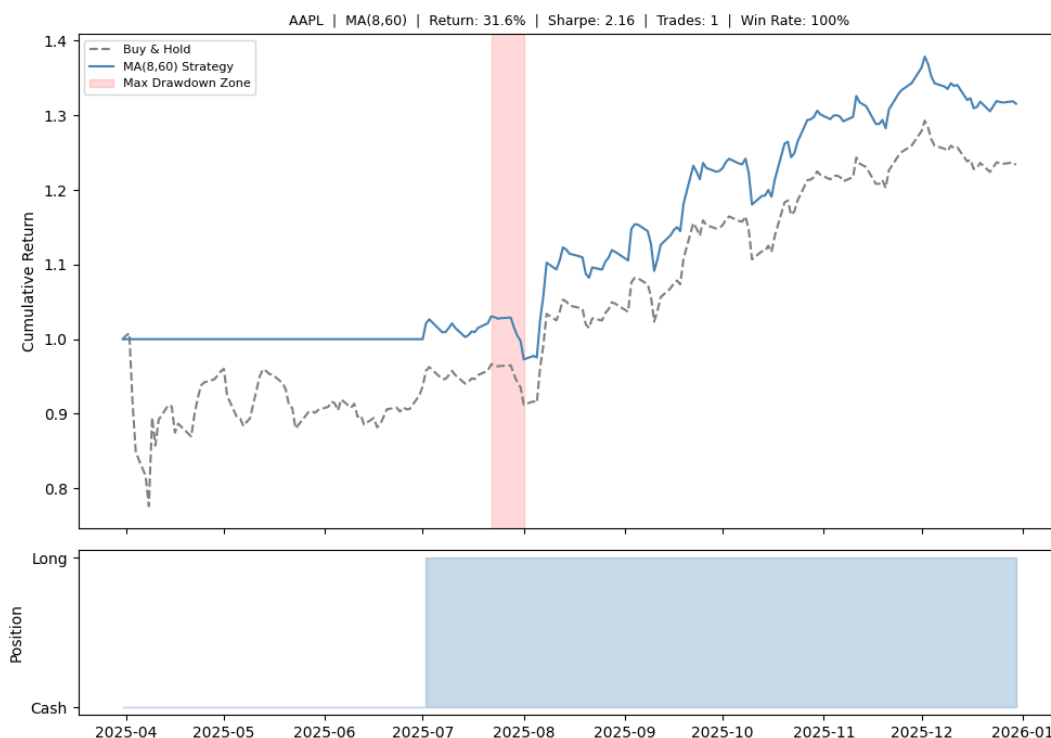


Figure 1: Strategy vs. buy-and-hold on AAPL (2025). The strategy entered once and held for 126 days without a stop-loss trigger. The lower panel shows position state (0 = cash, 1 = long).

4.1 The TSLA Case: Whipsaw in Equities

Figure 2 shows the TSLA result. The strategy entered four times, averaging just 17 days per trade, and returned -5.4% —while buy-and-hold gained approximately 80% over the same period. Two of the four exits were triggered by the stop-loss rather than by a Death Cross signal.

The mechanism is straightforward. TSLA’s day-to-day price swings are large enough to regularly breach the 5% stop threshold even during periods when the underlying trend is upward. The strategy exits what it interprets as a large adverse move, but the price recovers—so the strategy ends up selling near a local low. On the next Golden Cross signal it re-enters at a higher price than it sold. Repeating this cycle several times, each time paying a 0.1% commission, steadily drains the portfolio. This is the whipsaw effect.

4.2 The Bitcoin Case

Figure 3 confirms the same pattern at a more extreme level. Two of three positions were stopped out within weeks because normal BTC daily volatility routinely exceeds 5%. The strategy returned $+8.9\%$, which may appear acceptable in isolation, but this significantly underperformed buy-and-hold and was only achievable because the third position accumulated positive daily returns during its holding period before a Death Cross exit—an effect that appears in the cumulative equity curve but not in the trade-level win rate, for the reason noted in Section 2.5.

4.3 Root Cause: Fixed Thresholds in Variable-Volatility Environments

Comparing the three markets makes the underlying problem clear. A 5% single-day drop is unusual for AAPL and represents a meaningful signal. For TSLA, it happens several times per quarter. For Bitcoin, it can happen within a single afternoon.

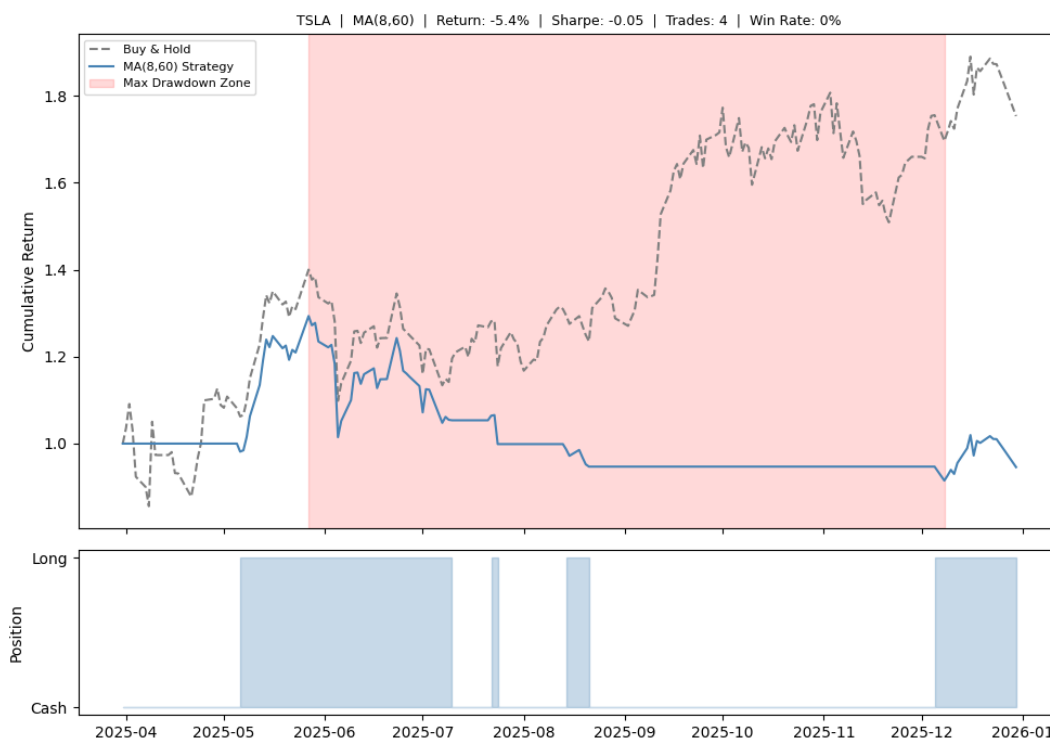


Figure 2: Strategy vs. buy-and-hold on TSLA (2025). Four trades averaging 17 days each, two stop-loss exits, total return -5.4% against approximately $+80\%$ buy-and-hold.

Using the same stop-loss percentage for all three assets is therefore not a consistent risk policy. It sets very different levels of sensitivity for each market: conservative enough for AAPL that it filters genuine noise, but so tight for TSLA and BTC that it reacts to routine fluctuations. The fixed stop-loss implicitly assumes volatility is constant across assets and over time, which is empirically false.

4.4 Future Direction

This failure points directly to what I need to learn next. Rather than a fixed percentage, I want to implement a stop-loss that adapts dynamically to each asset's current volatility. Specifically, I am interested in the **Average True Range (ATR)**—a measure of recent price range. Setting the stop at $2 \times \text{ATR}$ instead of a fixed 5% would automatically give volatile assets more room to fluctuate without triggering premature exits, while still exiting when a move is genuinely anomalous relative to recent behaviour.

This requires understanding how to measure and model time-varying volatility—topics I look forward to studying formally at university.

5 Conclusion

This project started as an attempt to build a simple trading strategy, but its real value came from what went wrong. The process of discovering, diagnosing, and quantifying the whipsaw effect gave me a clearer understanding of why quantitative finance requires statistical thinking, not just coding ability.

The key takeaways are:

1. Backtesting must be done carefully to avoid look-ahead bias; a day-by-day simulation loop is one way to enforce correct time ordering.

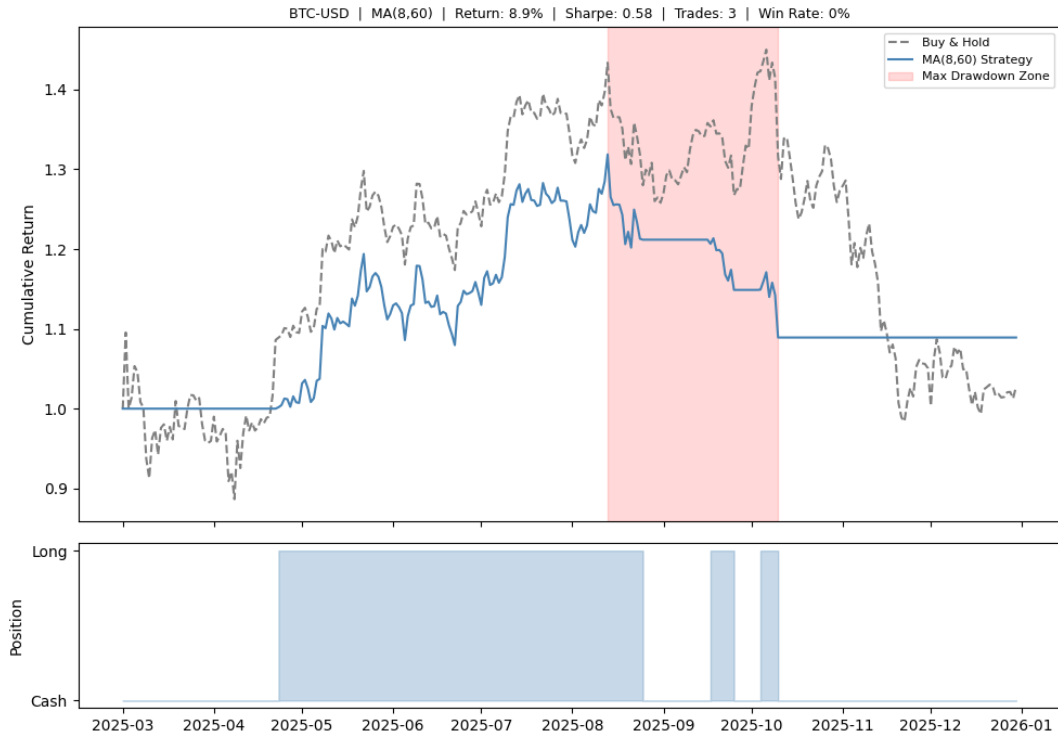


Figure 3: Strategy vs. buy-and-hold on BTC-USD (2025). Two of three positions triggered the stop-loss. The strategy returned +8.9% despite a 0% trade win rate, as daily gains during holding periods accumulated before each eventual stop-loss or exit.

2. Parameter optimisation on historical data is prone to overfitting; the AAPL results should be treated as an upper bound on likely real-world performance.
3. Risk management rules calibrated for one volatility regime can fail badly in another. TSLA's -5.4% strategy return versus $+80\%$ buy-and-hold is a concrete illustration.
4. Summary metrics like Sharpe Ratio alone can obscure the mechanism of failure; trade-level statistics (number of trades, average holding duration, stop-loss exit count) are often more informative diagnostics.

These lessons directly motivate my interest in studying statistics and mathematics at university, where I hope to develop the formal tools needed to address these challenges more rigorously.